

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Klastoets 2
VRAESTEL
Kopiereg voorbehou

Class Test 2
QUESTION PAPER
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Module EBN122
Elektrisiteit en Elektronika
September 2017

Module EBN122
Electricity and Electronics
September 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:
Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		9 + OCR	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 before the start of the session.

Do step 1 in the first 30 minutes of the test

Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN122C2**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2						k	Ω								
04	-	1																	

INSTRUKSIES

Doen stap 0 voor die begin van die sessie.

Doen stap 1 in die eerste 30 minute van die toets.

Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (-5 min to 0 min): Voltooi die voorblad van die OKH vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (30 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die "Assessment ID" is **2017EBN122C2**
- Sluit die korrekte eenheid vir alle antwoorde in. Maak gebruik van die toepaslike voorvoegsel.
- Gebruik 'n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

Voorbeelde

01	-	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2						k	Ω								
04	-	1																	

Question 01 [2]		Vraag 01 [2]	
Find the exclusive contribution of the independent voltage source to i_o .		Bepaal die eksklusiewe bydrae van die onafhanklike stroombron tot i_o .	
01	$i_o(20V)$	01	$i_o(20V)$

Let current source (2.5A) open circuit, then apply nodal analysis:

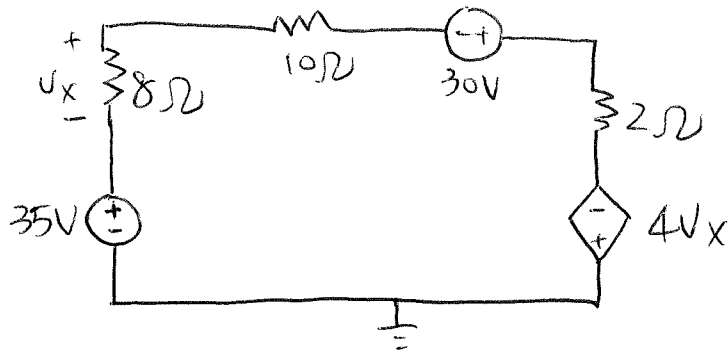
$$\frac{V-20}{8} + \frac{V}{3} = \frac{2V}{3} \Rightarrow \frac{V-20}{8} = \frac{V}{3}$$

$$3V - 60 = 8V \Rightarrow 5V = -60$$

$$V = -12(V) \Rightarrow i_o = \frac{V}{3} = -\frac{12}{3} = -4A$$

$$\underline{\hat{i}_o(20V) = -4A}$$

Question 02 [2]		Vraag 02 [2]	
Use source transformation to find v_x .		Gebruik brontransformase en bereken v_x .	
02	v_x	02	v_x



$$(35 + 30 + 4V_x) \cdot \frac{8}{20} = -V_x$$

$$65 + 4V_x = -2.5V_x$$

$$\Rightarrow -6.5V_x = 65$$

$$\Rightarrow \underline{V_x = -10 \text{ V}}$$

Question 03-04 [4]		Vraag 03-04 [2]	
Find the Thevenin equivalent (V_{Th} and R_{Th}) with respect to terminals $a-b$.		Bepaal die Thevenin ekwivalent (V_{Th} en R_{Th}) met verwysing tot terminale $a-b$.	
03	V_{Th}	03	V_{Th}
04	R_{Th}	04	R_{Th}

$$V_{Th} = 3V_x, \quad V_x = 3 \times (15 \parallel 10) = 18V$$

$$\Rightarrow \underline{V_{Th} = 18V \times 3 = 54V} \rightarrow$$

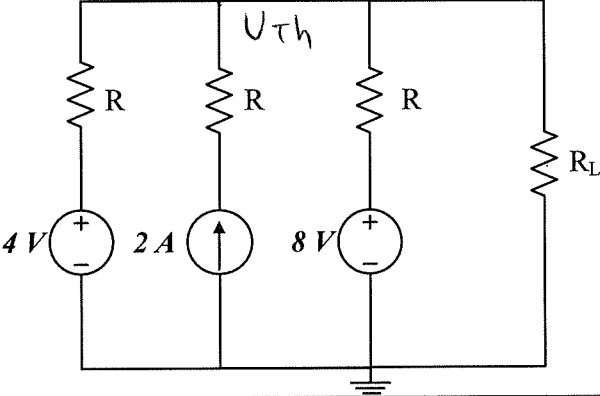
Get I_{sc} for R_{Th} , by nodal analysis:

$$3 = \frac{V'_x}{15} + \frac{V'_x}{10} + \frac{V'_x + 2V'_x}{30}$$

$$90 = 2V'_x + 3V'_x + 3V'_x$$

$$\Rightarrow V'_x = \frac{90}{8}V, \quad I_{sc} = \frac{V_x}{10} = \frac{9}{8}A$$

$$\Rightarrow \underline{R_{Th} = \frac{V_{Th}}{I_{sc}} = 54 \times \frac{8}{9} = 48\Omega} \rightarrow$$

Question 05 [2]		Vraag 05 [2]	
Find $P_{\max}$ of the load resistor R_L when $R = 10 \Omega$.			
05	$P_{\max}$	06	$P_{\max}$
			

$$R_{Th} = 0.5R = 5 \Omega$$

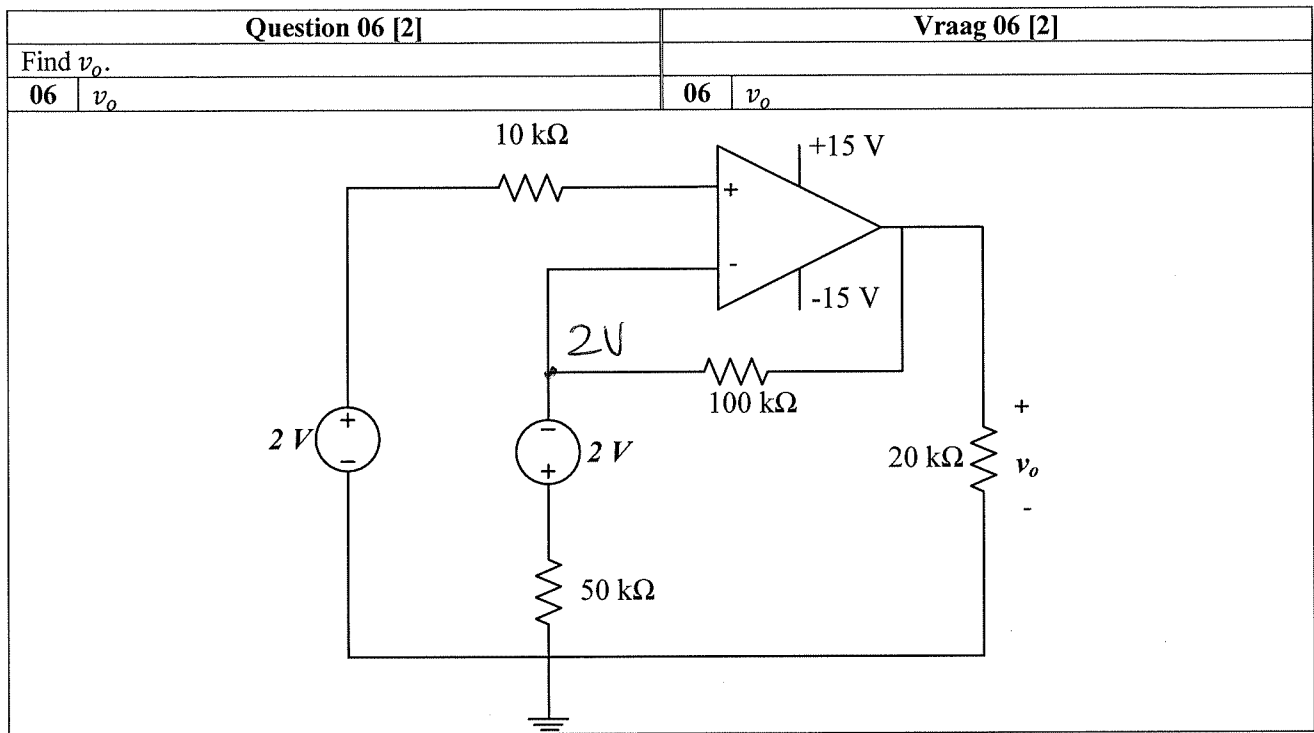
$$\frac{U_{Th} - 4}{10} + \frac{V_{Th} - 8}{10} = 2$$

$$\Rightarrow 2U_{Th} - 12 = 20$$

$$\Rightarrow U_{Th} = 16V$$

$$P_{\max} = \frac{U_{Th}^2}{4R_{Th}} = \frac{16^2}{4 \times 5} = 12.8W$$

_____D



$$\frac{2+2}{50K} + \frac{2-V_o}{100K} = 0$$

$$\Rightarrow 8 + 2 - V_o = 0$$

$$\Rightarrow V_o = 10V$$



Question 07-08 [2]		Vraag 07-08 [2]	
a) If $I_S = 0$ A and $V_S = 2$ V, calculate i_o .		a) Indien $I_S = 0$ A en $V_S = 2$ V, bereken i_o .	
b) If $I_S = 0.2$ mA and $V_S = 0$ V, calculate v_o .		b) Indien $I_S = 0.2$ mA en $V_S = 0$ V, bereken v_o .	
07	i_o	07	i_o
08	v_o	08	v_o

a) $I_S = 0$, non-inverting op-amp, then

$$V_o = V_S \left(1 + \frac{50\text{k}}{20\text{k}} \right) = 2 \times 3.5 = 7\text{ V};$$

$$i_o = \frac{V_o}{2\text{k}\Omega} = 3.5\text{ mA}$$

b) $V_S = 0$, inverting op-amp after source transformation of I_S :

$$V_o = - (0.2 \times 10^{-3} \times 10 \times 10^3) \times \frac{50\text{k}}{20\text{k}} = -5\text{ V}$$

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	7	E	B	N	1	2	2	C	2		
	Answer											Unit		

01	$i_{0(20V)} = -4$	A
02	$v_x = -10$	V
03	$V_{Th} = 54$	V
04	$R_{Th} = 48$	Ω
05	$P_{max} = 12.8$	W
06	$v_o = 10$	V
07	$i_o = 3.5$	mA
08	$v_o = -5$	V